

Self-Attention Sliding Window Enhanced Canonical Correlation Analysis for Incipient Fault Detection in Dynamic Industrial Processes

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Abstract—Thanks to its excellent nonlinear representation ability, deep neural network (DNN) has been designed to aid canonical correlation analysis (CCA) for stable kernel representation. These DNN-aided CCA methods make use of correlation as the optimization objective and exploit its change to distinguish system states. However, the maximum correlation-based training strategy lacks robustness to effectively tackle system false alarms caused by changes in operating conditions. To this end, this work proposes a self-attention sliding window enhanced CCA (SaECCA) for incipient fault detection in dynamic industrial processes. The main novelties of this work include the following: first, with the aid of DNNs, an adaptive weighting mechanism with self-attention is developed to amplify the incipient fault information; second, residual estimation-based, a novel and robust optimization objective for nonlinear CCA is formulated; third, an SaECCA-based fault detection algorithm is designed, whose convergence and detectability are illustrated via theoretical analysis. Studies on a three-tank system simulation and a multiphase flow industrial process are presented to verify the effectiveness of the proposed SaECCA method.

Index Terms—Canonical correlation analysis (CCA), dynamic industrial processes, incipient fault detection (FD), self-attention sliding window.

I. INTRODUCTION

THE rapid advancement of information technology has made modern industrial processes increasingly automated and large-scale. This evolution not only brings huge economic benefits, but also further exacerbates the risk of system failure [1]. For the industrial process, an incipient fault usually

has no obvious symptoms and will progress into a system-level failure if it cannot be discovered in time. To this end, developing a reliable fault detection (FD) strategy that timely identifies early anomalies and performs the necessary maintenance is essential to ensuring equipment safety and product quality [2], [3], [4], [5].

Thanks to the massive deployment of advanced sensors, data-driven FD techniques [5], [6], [7] have attracted special attention over the recent two decades due to their simplicity and interpretability. For example, principal component analysis (PCA) [6], partial least squares [8], and canonical correlation analysis (CCA) [9] are typical representatives of this field. Among these methods, CCA seeks the optimal projection direction by maximizing the linear correlation between process and output data [10]. Note that CCA is a data-driven FD method based on residual signals, and its parameters can be obtained directly from constructing the residual generator without identifying the state-space model of the system. This feature enables CCA to perform rapid anomaly detection even with unknown system models [11], [12], [13].

Building on this, Chen et al. [10] proposed a linear CCA-based FD algorithm for static industrial systems. However, linear CCA-based FD methods assume independent and identically distributed data [11] and strictly linear correlations between measurement signals. For complex industrial systems with dynamic and nonlinear characteristics, these assumptions do not hold, thus directly applying linear CCA cannot be optimal. To address dynamic characterization, Hu et al. [13] proposed a sparse dynamic CCA by introducing sparsity in feature extraction, thereby enhancing both interpretability and computational efficiency. In [14], a multiblock dynamic enhanced CCA was developed for monitoring the combustion status of the waste incinerator.

Except for the dynamic characterization [13], CCA has also been extended to address nonlinear relationships. Drawing on the idea that a nonlinear process can be approximated by multiple local CCAs, a locally weighted CCA method was developed in [15]. In [16], a dynamic concurrent kernel CCA was developed to capture nonlinear dynamics between strip thickness and process variables during cold-rolled steel annealing. In addition, by establishing regression relationships between process and quality variables, a quality-related FD method called kernel PCA-CCA was developed in [17].

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Recently, deep neural network (DNN) has been widely used for FD in nonlinear systems [18], owing to their powerful nonlinear learning ability. With the aid of DNN, various DNN-based CCA methods have been developed. Andrew et al. [19] first achieved higher correlation results in their pioneering work by using DNN to learn the nonlinear transformations of two data views. Different from the kernel-based methods, DNN-aided CCA avoids inner product computations, and its parameterization mechanism allows training time to scale with data size. Motivated by this conclusion, Chen et al. [20] proposed a DNN-CCA framework for dynamic process monitoring and FD purpose. In addition, Chen et al. [12] reformulated the objective function of nonlinear CCA, providing its generalized solution and optimal diagnostic performance via theoretical analysis. To enhance the extraction of spatio-temporal features from the data, an improved two-stream neural network CCA (TsnnCCA) was proposed in [21].

The above studies have indicated that the existing CCA-based FD methods can effectively detect faults in nonlinear and dynamic processes. However, most of these methods focused on detecting abrupt faults, and satisfying FD performance usually cannot be achieved when confronted with incipient faults exhibiting small amplitudes. To make the detection index more sensitive for incipient faults, Chen et al. [22] integrated Kullback–Leibler divergence (KLD) into independent component analysis (ICA) framework, and developed a ICA-KLD-based FD method. Inspired by the literature [4], Ji et al. [23] proposed a canonical variable residual statistics analysis method (CVRSA). The CVRSA is designed to improve the ability of canonical variable residuals for incipient faults by creating a sliding window and constructing a detection index based on the Mahalanobis distance. In addition, Chen et al. [24] developed a probability-relevant PCA (PRPCA) to tackle the crowding problem caused by incipient faults from the perspective of probability distribution.

Up to now, these latest methods provide a certain extent of improvement in the detection ability for incipient faults, but still do not utilize the advantages of DNNs in feature extraction and nonlinear representation. In particular, the research on how DNNs can be used to aid CCA in improving the detectability of incipient faults is still in the preliminary stage. To this end, building on existing research, a self-attention sliding window enhanced CCA (SaECCA) method is proposed to achieve robust and effective detection for incipient faults. The main contributions of this work are summarized as follows.

- 1) A SaECCA-based FD framework is designed. With the advantages of DNN in nonlinear mapping, the generated nonlinear projections achieve maximum correlation.
- 2) For the time-varying nature of industrial processes, the weighting strategy with self-attention sliding window is developed to improve the detectability of incipient faults.
- 3) Different from the maximum correlation-based training strategy, a novel and robust optimization objective is formulated for DNN-aided CCA. Its optimal solution is realized based on the minimized estimation errors.

The rest of the article is organized as follows. Section II introduces CCA-based FD method and the problem to be solved.

In Section III, a SaECCA-based FD algorithm is developed. The convergence and FD ability of SaECCA are elaborated in Section IV. Experimental study on a three-tank system (TTS) simulation and a multiphase flow (MPF) industrial process is given in Section V. Finally, Section VI concludes this article.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. CCA-Based FD Method

Given a linear process with m process variables $\mathbf{u}_t \in \mathbb{R}^m$ and l output variables $\mathbf{y}_t \in \mathbb{R}^l$. Collect N such samples into the process matrix $\mathbf{U}$ and the output matrix $\mathbf{Y}$, that

$$\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_N] \in \mathbb{R}^{m \times N} \quad (1)$$

$$\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_N] \in \mathbb{R}^{l \times N}. \quad (2)$$

Let $(\mathbf{U}, \mathbf{Y})$ be zero-mean and normal distribution. Denote their covariance matrices by Σ_u , Σ_y , and Σ_{uy} . The linear CCA aims to maximize the correlation of $(\mathbf{J}_c^T \mathbf{u}_t, \mathbf{L}_c^T \mathbf{y}_t)$ by a pair of linear projections [25]

$$\begin{aligned} (\mathbf{J}_c^*, \mathbf{L}_c^*) &= \arg \max_{\mathbf{J}_c, \mathbf{L}_c} \rho(\mathbf{J}_c^T \mathbf{u}_t, \mathbf{L}_c^T \mathbf{y}_t) \\ \text{s.t. } \mathbf{J}_c^T \Sigma_u \mathbf{J}_c &= \mathbf{L}_c^T \Sigma_y \mathbf{L}_c = \mathbf{I} \end{aligned} \quad (3)$$

where $(\mathbf{J}_c^*, \mathbf{L}_c^*)$ are optimal projections, ρ is a correlation metric, and $\mathbf{I}$ is an identity matrix. For (3), a popular way is to perform a singular value decomposition (SVD) [15]

$$\Sigma_u^{-1/2} \Sigma_{uy} \Sigma_y^{-1/2} = \mathbf{\Gamma} \mathbf{\Lambda} \mathbf{\Delta}^T \quad (4)$$

where $\mathbf{\Gamma} = [\gamma_1, \dots, \gamma_\kappa]$, $\mathbf{\Lambda} = \text{diag}(\sigma_1, \dots, \sigma_\kappa)$, and $\mathbf{\Delta} = [v_1, \dots, v_\kappa]$, and $\kappa = \text{rank}(\mathbf{\Lambda}) \leq \min\{m, l\}$ is the number of nonzero singular values in $\mathbf{\Lambda}$.

Let $\mathbf{J}_c = \Sigma_u^{-1/2} \mathbf{\Gamma}$ and $\mathbf{L}_c = \Sigma_y^{-1/2} \mathbf{\Delta}$, the linear CCA-based residual generators are defined for the FD purpose

$$\mathbf{r}_u = \mathbf{J}_c^T \mathbf{u}_t - \mathbf{\Lambda} \mathbf{L}_c^T \mathbf{y}_t \quad (5)$$

$$\mathbf{r}_y = \mathbf{L}_c^T \mathbf{y}_t - \mathbf{\Lambda} \mathbf{J}_c^T \mathbf{u}_t. \quad (6)$$

As described in [20], CCA-based methods define a residual index $T^2 = \mathbf{r}^T (\mathbf{I} - \mathbf{\Lambda}^2)^{-1} \mathbf{r}$, that is, $\mathbf{r} = \mathbf{r}_u$ or $\mathbf{r}_y$, alongside a corresponding threshold J_{th} for FD. Once $T^2 > J_{th}$, the system is deduced to be faulty.

B. Problem Statements

Consider a feedback control system $\mathcal{S}_0$ with the dataset $\mathbf{Z} = [\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_N]$. Under steady-state conditions, $\mathcal{S}_0$ can be defined as [12]

$$\mathcal{S}_0 : \mathbf{z}_t = \begin{bmatrix} \mathbf{u}_t \\ \mathbf{y}_t \end{bmatrix}, \quad \mathbf{y}_t = \varphi(\mathbf{u}_t) \quad (7)$$

where $\mathbf{z}_t$ contains all variables and $\varphi : U \rightarrow Y$ is a nonlinear operator that describes the mapping relationship between the process variables $\mathbf{u}_t$ and output variables $\mathbf{y}_t$. In addition, an equivalent description of $\mathcal{S}_0$ in (7) can be derived by using SVD for Σ_u and Σ_y , as elaborated in [26].

For $\mathcal{S}_0$ in (7), its performance are affected by the actuator faults $\mathcal{S}_{f_u}$ and sensor faults $\mathcal{S}_{f_y}$ [27]. With both faults, the feedback

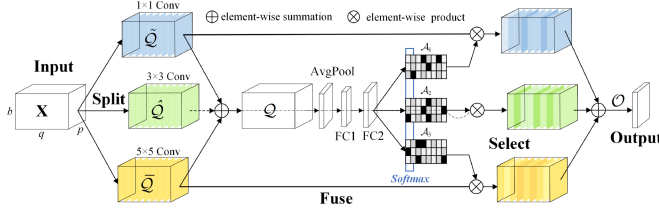


Fig. 1. Schematic of SKConv-based nonlinear representations.

control system becomes

$$\mathcal{S}_{f_u} : \mathbf{z}_f = \begin{bmatrix} \mathbf{u}_t^f \\ \mathbf{y}_t^f \end{bmatrix} \text{ and } \mathcal{S}_{f_y} : \mathbf{z}_f = \begin{bmatrix} \mathbf{u}_t^f \\ \mathbf{y}_t^f \end{bmatrix} \quad (8)$$

where

$$\mathbf{u}_t^f = \mathbf{u}_t + \Xi_u f_u, \quad \mathbf{y}_t^f = \mathbf{y}_t + \Xi_y f_y. \quad (9)$$

Here, $\Xi_u f_u$ and $\Xi_y f_y$ are the actuation and sensor faulty parts, respectively; Ξ_u and Ξ_y are the fault direction; f_u and f_y are the fault magnitude.

If $\mathcal{S}_0$ is affected by actuator faults $\mathcal{S}_{f_u}$, the $\mathbf{r}_u$ can be derived by integrating $\mathbf{u}_t^f$ in (9) into (5), that is:

$$\mathbf{r}_u = \underbrace{\mathbf{J}^T \mathbf{u}_t - \mathbf{A} \mathbf{L}^T \mathbf{y}_t}_{\mathbf{r}_n} + \underbrace{\mathbf{J}^T \Xi_u f_u}_{\mathbf{r}_f}. \quad (10)$$

In (10), $\mathbf{r}_n$ and $\mathbf{r}_f$ are the residuals of the normal and faulty parts. Similarly, we can derive the residual generator $\mathbf{r}_y$:

$$\mathbf{r}_y = \underbrace{\mathbf{L}^T \mathbf{y}_t - \mathbf{A} \mathbf{J}^T \mathbf{u}_t}_{\mathbf{r}_n} + \underbrace{\mathbf{L}^T \Xi_y f_y}_{\mathbf{r}_f}. \quad (11)$$

For FD purpose, $\mathbf{r}_n$ should be minimized to maintain sensitivity to faults, while $\mathbf{r}_f$ should be maximized to enhance faults detectability. In practice, $\mathbf{r}_n$ is driven by operating conditions and noises so that it cannot always equal to 0, and $\mathbf{r}_f$ increases with the intensity of f_u and f_y .

As the magnitudes f_u and f_y for the incipient faults are quite slight, a popular strategy is to introduce the moving average (MA) with a sliding window [28]. Given a window width w , the MA is described by:

$$\text{MA} : \bar{\mathbf{u}}_t = \frac{1}{w} \sum_{p=0}^{w-1} \mathbf{u}_{t-p}, \quad \bar{\mathbf{y}}_t = \frac{1}{w} \sum_{p=0}^{w-1} \mathbf{y}_{t-p}. \quad (12)$$

In (12), MA enhances the detection ability of $\mathbf{r}_u$ and $\mathbf{r}_y$ for the incipient faults by averaging current and past $w-1$ samples, with equal weights assigned to all instants. However, this weighting mechanism cannot be suitable because industrial processes are time-varying. Moreover, existing DNN-aided CCA methods mainly employ a correlation-based optimization paradigm, but the residual changes caused by working condition shifts within the dynamic process can weaken the robustness of such training scheme, increasing false alarm generation. Based on this, the left problem is how to design an adaptive weighting mechanism and a robust optimization objective for the DNN-aided CCA methods. In brief, the objectives of this work include the following.

- 1) To design two DNNs with self-attention sliding window to replace the kernel function for stable kernel representation (SKR) and adaptively adjust the weights.
- 2) To formulate a more robust optimization objective for DNN-aided CCA methods by equivalently transforms maximize correlation into minimize residuals.
- 3) To theoretically analyze the convergence and FD ability of the proposed SaECCA algorithm.

III. SAECCA-BASED FD

A. Nonlinear Representations Based on DNNs

Benefiting from its excellent nonlinear learning representation ability, DNN is used to replace the kernel function ϕ for explicit nonlinear mapping. Given arbitrary two DNNs with the same structure, the following SKR can be obtained:

$$\underline{\mathbf{u}} = \mathcal{H}(\mathbf{u}, \Theta_{\mathcal{H}}) \quad (13)$$

$$\underline{\mathbf{y}} = \mathcal{G}(\mathbf{y}, \Theta_{\mathcal{G}}) \quad (14)$$

where $\mathcal{H}(\Theta_{\mathcal{H}})$ and $\mathcal{G}(\Theta_{\mathcal{G}})$ denote nonlinear operators with parameters $\Theta_{\mathcal{H}} := (\mathbf{W}_{\mathcal{H}}, \mathbf{b}_{\mathcal{H}})$ and $\Theta_{\mathcal{G}} := (\mathbf{W}_{\mathcal{G}}, \mathbf{b}_{\mathcal{G}})$. $\mathbf{W}$ and $\mathbf{b}$ are the network weights and bias terms, respectively, and $\underline{\mathbf{u}}$ and $\underline{\mathbf{y}}$ are DNNs outputs.

It is a fact that for the nonlinear operators $\mathcal{H}(\Theta_{\mathcal{H}})$ and $\mathcal{G}(\Theta_{\mathcal{G}})$ defined in (13) and (14), they can always generate the corresponding suspected spaces [12], $\mathcal{H}(\Theta_{\mathcal{H}}) : \mathcal{U} \rightarrow \underline{\mathcal{U}}$ and $\mathcal{G}(\Theta_{\mathcal{G}}) : \mathcal{Y} \rightarrow \underline{\mathcal{Y}}$. To obtain all features (i.e., representations) in $\underline{\mathcal{U}}$ and $\underline{\mathcal{Y}}$, the first task is to select a suitable DNN model. Among the various DNNs, convolutional neural network (CNN) is one of the most successful network architectures, which achieves SKR via simple convolutional operations and nonlinear activation functions instead of complex kernel computations [16]. Based on this, selective kernel convolution (SKConv) is employed to aid CCA for SKR, which is a CNN extension [29]. The SKConv schematic with three branches is illustrated in Fig. 1. The SKConv is primarily realized via three operations: *split*, *fuse*, and *select* [29]. The details are provided below.

1) *Split*: Given an input $\mathbf{X} \in \mathbb{R}^{b \times p \times q}$, we first perform the following transformation:

$$\tilde{\mathcal{F}} : \mathbf{X} \rightarrow \tilde{\mathbf{Q}} \in \mathbb{R}^{b \times p \times q} \quad (15)$$

$$\hat{\mathcal{F}} : \mathbf{X} \rightarrow \hat{\mathbf{Q}} \in \mathbb{R}^{b \times p \times q} \quad (16)$$

$$\bar{\mathcal{F}} : \mathbf{X} \rightarrow \bar{\mathbf{Q}} \in \mathbb{R}^{b \times p \times q} \quad (17)$$

where $\tilde{\mathcal{F}}$, $\hat{\mathcal{F}}$, and $\bar{\mathcal{F}}$ denote $1 \times 1 \text{ Conv}$, $3 \times 3 \text{ Conv}$, and $5 \times 5 \text{ Conv}$ operations, respectively; b , p , and q denote batch size, time lags, and variable dimensions, respectively. Note that $\tilde{\mathcal{F}}$, $\hat{\mathcal{F}}$, and $\bar{\mathcal{F}}$ are each sequentially composed of efficient dilated convolutions [29], batch normalization [30], and rectified linear unit (ReLU) [20] activation function.

Remark 1: Different from the matrix-based data representation for the dynamic CCA in [13]. The tensor-based data representation in DNNs is inherently dynamic [20], and its inputs retain the original structure of the data, which allows for a more effective capture of temporal dynamics.

2) *Fuse*: To aggregate multiscale features, a basic idea is to use gates to control branches that carry different scales of information. To achieve this, the results of $\tilde{\mathcal{F}}$, $\hat{\mathcal{F}}$, and $\bar{\mathcal{F}}$ are fused via an element-wise summation:

$$\mathcal{Q} = \tilde{\mathcal{Q}} + \hat{\mathcal{Q}} + \bar{\mathcal{Q}}. \quad (18)$$

Then, the temporal statistics $\mathcal{P} \in \mathbb{R}^{b \times p}$ can be obtained by simply using average pooling. Specifically, the temporal statistics $\mathcal{P}_c$ at the c th window are calculated by shrinking $\mathcal{Q}$ across variable dimensions q :

$$\mathcal{P}_c = \text{AvgPool}(\mathcal{Q}_c) = \frac{1}{q} \sum_{v=1}^q \mathcal{Q}_c(v). \quad (19)$$

Further, the temporal statistics $\mathcal{P}_c$ are compressed and reconstructed through two fully connected (FC) layers, as shown in Fig. 1. Here, FC1 is used to guide feature selection and extract key temporal information, while FC2 is employed to restore the sequence structure and enhance the representational ability of the model:

$$\mathcal{D}_c = \mathcal{F}_{fc}(\mathcal{P}_c) = \delta(\mathcal{B}(\mathbf{W}_c^1 \mathcal{P}_c + \mathbf{b}_c^1)) \quad (20)$$

$$\mathcal{Z}_c = \mathcal{F}_{fc}(\mathcal{D}_c) = \delta(\mathcal{B}(\mathbf{W}_c^2 \mathcal{D}_c + \mathbf{b}_c^2)) \quad (21)$$

where $\delta(\cdot)$ and $\mathcal{B}(\cdot)$ are ReLU function and Batch Normalization [30]; $\mathbf{W}_c^h$ and $\mathbf{b}_c^h$ denote the weights and bias at the h th FC layer, $h = 1, 2$; $\mathcal{D}_c \in \mathbb{R}^{1 \times d}$ and $\mathcal{Z}_c \in \mathbb{R}^{1 \times p}$ denote FC1 and FC2 outputs, respectively.

3) *Select*: Guided by the temporal information $\mathcal{Z}$, the soft attention is employed to adaptively select the cross-time relationships. Specifically, the soft attention weights of $\tilde{\mathcal{Q}}$, $\hat{\mathcal{Q}}$, and $\bar{\mathcal{Q}}$ can be obtained by the following softmax operation:

$$\tilde{\mathcal{A}}_c = \frac{e^{\tilde{\mathcal{A}}\mathcal{Z}}}{e^{\tilde{\mathcal{A}}\mathcal{Z}} + e^{\hat{\mathcal{A}}\mathcal{Z}} + e^{\bar{\mathcal{A}}\mathcal{Z}}} \quad (22)$$

$$\hat{\mathcal{A}}_c = \frac{e^{\hat{\mathcal{A}}\mathcal{Z}}}{e^{\tilde{\mathcal{A}}\mathcal{Z}} + e^{\hat{\mathcal{A}}\mathcal{Z}} + e^{\bar{\mathcal{A}}\mathcal{Z}}} \quad (23)$$

$$\bar{\mathcal{A}}_c = \frac{e^{\bar{\mathcal{A}}\mathcal{Z}}}{e^{\tilde{\mathcal{A}}\mathcal{Z}} + e^{\hat{\mathcal{A}}\mathcal{Z}} + e^{\bar{\mathcal{A}}\mathcal{Z}}} \quad (24)$$

where $\tilde{\mathcal{A}}$, $\hat{\mathcal{A}}$, and $\bar{\mathcal{A}} \in \mathbb{R}^{b \times p}$ represent the soft attention matrices for $\tilde{\mathcal{Q}}$, $\hat{\mathcal{Q}}$, and $\bar{\mathcal{Q}}$, respectively, or referred to as the weight matrices. Note that the weight vector $\mathcal{A}_c \in \mathbb{R}^{1 \times p}$ for the c th window is dynamically learned through the soft self-attention mechanism, and it satisfies $\tilde{\mathcal{A}}_c + \hat{\mathcal{A}}_c + \bar{\mathcal{A}}_c = 1$.

In the case of three branches, the element-wise product and summation are applied to the fused features $\mathcal{Q}$ to obtain the SKR corresponding to the input $\mathbf{X}$:

$$\mathcal{O}_c = \frac{1}{p} \sum_{t=1}^p \left(\tilde{\mathcal{A}}_c \tilde{\mathcal{Q}}_c(t) + \hat{\mathcal{A}}_c \hat{\mathcal{Q}}_c(t) + \bar{\mathcal{A}}_c \bar{\mathcal{Q}}_c(t) \right) \quad (25)$$

where $\mathcal{O} = [\mathcal{O}_1^T, \mathcal{O}_2^T, \dots, \mathcal{O}_b^T]^T \in \mathbb{R}^{b \times m}$ denotes the SKR of the input $\mathbf{X}$.

Remark 2: It is worth noting that for the SKConv with three branches described in Section III-A, it can be easily derived to any branches by extending (18) and (22)–(24).

Remark 3: Compared with MA, the softmax operation in (22)–(24) constantly adjusts the soft attention weights $\tilde{\mathcal{A}}$, $\hat{\mathcal{A}}$, and $\bar{\mathcal{A}}$ based on temporal information $\mathcal{Z}$, and subsequently performs the element-wise product and summation.

B. Reformulation of Nonlinear CCA Objective Function

As described in Section II-A, the generalized objective function of CCA can be summarized in [19]

$$\mathcal{L}_{\text{corr}} = \max \rho(\underline{\mathbf{u}}, \underline{\mathbf{y}}). \quad (26)$$

In practice, because of the existence of noise disturbances in the feedback system $\mathcal{S}_0$, the generalized objective function in (26) satisfies:

$$0 < \rho(\underline{\mathbf{u}}, \underline{\mathbf{y}}) < \kappa. \quad (27)$$

As pointed out in [12], (27) provides a feasible metric for evaluating whether the operators $\mathcal{H}(\Theta_{\mathcal{H}})$ and $\mathcal{G}(\Theta_{\mathcal{G}})$ are optimal. In addition, the DNNs aim to minimize the loss, while (26) is a maximization problem. Hence, the objective function of DNN-aided CCA is modified as follows [20]:

$$\begin{aligned} \mathcal{L}_{\text{corr}} &= \kappa - \rho(\underline{\mathbf{u}}, \underline{\mathbf{y}}) \\ &= \kappa - \text{Tr} \left(\left| \Sigma_{\underline{\mathbf{u}}}^{-1/2} \Sigma_{\underline{\mathbf{u}}\underline{\mathbf{y}}} \Sigma_{\underline{\mathbf{y}}}^{-1/2} \right| \right) \end{aligned} \quad (28)$$

where $\text{Tr}(\cdot)$ and $|\cdot|$ are the trace and absolute value operator, $\Sigma_{\underline{\mathbf{u}}}$, respectively, $\Sigma_{\underline{\mathbf{u}}\underline{\mathbf{y}}}$, and $\Sigma_{\underline{\mathbf{y}}}$ are the covariance and cross-covariance of the $\underline{\mathbf{u}}$ and $\underline{\mathbf{y}}$, respectively, $\underline{\mathbf{u}}$ and $\underline{\mathbf{y}}$ are mean centered.

In fact, the optimal network operators $\mathcal{H}(\Theta_{\mathcal{H}}^*) : U \rightarrow U^*$ and $\mathcal{G}(\Theta_{\mathcal{G}}^*) : Y \rightarrow Y^*$ can be generated by minimizing the error between κ and the total correlation ρ . To achieve the optimal network operators $\mathcal{H}(\Theta_{\mathcal{H}}^*)$ and $\mathcal{G}(\Theta_{\mathcal{G}}^*)$, two SKConv modules with same structures are used to optimize $\mathcal{L}_{\text{corr}}$ in (28). End with the training of two SKConv modules, the network parameters $\Theta_{\mathcal{H}}^*$ and $\Theta_{\mathcal{G}}^*$ are retained. Then, performing SVD on $\Sigma_{\underline{\mathbf{u}}}^{-1/2} \Sigma_{\underline{\mathbf{u}}\underline{\mathbf{y}}} \Sigma_{\underline{\mathbf{y}}}^{-1/2}$, the projection directions ($\mathbf{J}_{\mathcal{H}}$, $\mathbf{L}_{\mathcal{G}}$) and the eigenvalues matrix Λ_{κ} can be obtained. Based on this, the residual generators $\mathbf{r}_{\underline{\mathbf{u}}}$ and $\mathbf{r}_{\underline{\mathbf{y}}}$, which are similar to (5) and (6), can be formulated:

$$\mathbf{r}_{\underline{\mathbf{u}}} = \mathbf{J}_{\mathcal{H}}^T \underline{\mathbf{u}} - \Lambda_{\kappa} \mathbf{L}_{\mathcal{G}}^T \underline{\mathbf{y}} \quad (29)$$

$$\mathbf{r}_{\underline{\mathbf{y}}} = \mathbf{L}_{\mathcal{G}}^T \underline{\mathbf{y}} - \Lambda_{\kappa} \mathbf{J}_{\mathcal{H}}^T \underline{\mathbf{u}}. \quad (30)$$

Note that the residual generators $\mathbf{r}_{\underline{\mathbf{u}}}$ and $\mathbf{r}_{\underline{\mathbf{y}}}$ distinguish the system states by constructing correlation-based causal dependencies, which the generated residual signals that decrease as $\rho(\underline{\mathbf{u}}, \underline{\mathbf{y}})$ increases. However, system condition variations and noises can easily weaken this correlation, causing the residual signals to increase abruptly, even exceeding the corresponding thresholds. In addition, because incipient fault symptoms are not obvious, it is difficult for the maximum correlation-based training strategy to detect such faults through correlation changes. Therefore, the objective function $\mathcal{L}_{\text{corr}}$ in (28) cannot be reliable for incipient FD purpose. Given that CCA-based FD methods inherently rely on the residual changes between $\underline{\mathbf{u}}$ and $\underline{\mathbf{y}}$ to identify system faults [10]. The network parameters of the

proposed SaECCA are optimized by:

$$(\Theta_{\mathcal{H}}^*, \Theta_{\mathcal{G}}^*) = \arg \min \|\mathbf{r}_{\underline{u}}\|_2^2 + \|\mathbf{r}_{\underline{y}}\|_2^2. \quad (31)$$

Further, the optimization objective (also be called loss function) $\mathcal{L}_r$ of SaECCA is specifically reformulated as follows:

$$\begin{aligned} \min_{\Theta_{\mathcal{H}}, \Theta_{\mathcal{G}}} \quad & \|\mathbf{r}_{\underline{u}}\|_2^2 + \|\mathbf{r}_{\underline{y}}\|_2^2 \\ \text{s.t.} \quad & \mathbf{J}_{\mathcal{H}}^T (\mathbf{u}\mathbf{u}^T + r_1 \mathbf{I}) \mathbf{J}_{\mathcal{H}} = \mathbf{I} \\ & \mathbf{L}_{\mathcal{G}}^T (\mathbf{y}\mathbf{y}^T + r_2 \mathbf{I}) \mathbf{L}_{\mathcal{G}} = \mathbf{I} \end{aligned} \quad (32)$$

where $r_1 > 0$ and $r_2 > 0$ are two regularization parameters with small real values. They are introduced to avoid the numerical problem arising from the inverse of the covariance matrices [21].

Theorem 1: To achieve the optimal network operators $\mathcal{H}(\Theta_{\mathcal{H}}^*) : U \rightarrow \underline{U}^*$ and $\mathcal{G}(\Theta_{\mathcal{G}}^*) : Y \rightarrow \underline{Y}^*$, $\mathcal{L}_{\text{corr}}$ aligns with the objective function defined in (32), that is

$$\max \rho(\mathbf{u}, \mathbf{y}) \iff \min \|\mathbf{r}_{\underline{u}}\|_2^2 + \|\mathbf{r}_{\underline{y}}\|_2^2. \quad (33)$$

Proof: For the residual generator $\mathbf{r}_{\underline{u}}$ in (29), its covariance can be represented as

$$\begin{aligned} \mathbb{E}[\mathbf{r}_{\underline{u}}\mathbf{r}_{\underline{u}}^T] &= \mathbb{E}[\mathbf{J}_{\mathcal{H}}^T \Sigma_{\underline{u}} \mathbf{J}_{\mathcal{H}}] + \mathbb{E}[\Lambda_{\kappa} \mathbf{L}_{\mathcal{G}}^T \Sigma_{\underline{y}} \mathbf{L}_{\mathcal{G}} \Lambda_{\kappa}^T] \\ &\quad - \mathbb{E}[\mathbf{J}_{\mathcal{H}}^T \Sigma_{\underline{u}\underline{y}} \mathbf{L}_{\mathcal{G}} \Lambda_{\kappa}^T] - \mathbb{E}[\Lambda_{\kappa} \mathbf{L}_{\mathcal{G}}^T \Sigma_{\underline{y}\underline{u}} \mathbf{J}_{\mathcal{H}}] \end{aligned} \quad (34)$$

where $\mathbb{E}[\cdot]$ is expectation operator.

According to the orthogonality of SVD and $\mathbf{J}_{\mathcal{H}}^T \Sigma_{\underline{u}\underline{y}} \mathbf{L}_{\mathcal{G}} = \Lambda_{\kappa}$ in [14], it can be verified that [31]

$$\begin{aligned} \mathbb{E}[\mathbf{J}_{\mathcal{H}}^T \Sigma_{\underline{u}} \mathbf{J}_{\mathcal{H}}] &= (1 - r_1) \mathbf{I}_{\kappa} \\ \mathbb{E}[\Lambda_{\kappa} \mathbf{L}_{\mathcal{G}}^T \Sigma_{\underline{y}} \mathbf{L}_{\mathcal{G}} \Lambda_{\kappa}^T] &= (1 - r_2) \Lambda_{\kappa} \Lambda_{\kappa}^T \\ \mathbb{E}[\mathbf{J}_{\mathcal{H}}^T \Sigma_{\underline{u}\underline{y}} \mathbf{L}_{\mathcal{G}} \Lambda_{\kappa}^T] &= \Lambda_{\kappa} \Lambda_{\kappa}^T \\ \mathbb{E}[\Lambda_{\kappa} \mathbf{L}_{\mathcal{G}}^T \Sigma_{\underline{y}\underline{u}} \mathbf{J}_{\mathcal{H}}] &= \Lambda_{\kappa} \Lambda_{\kappa}^T. \end{aligned} \quad (35)$$

Then, $\mathbb{E}[\mathbf{r}_{\underline{u}}\mathbf{r}_{\underline{u}}^T]$ in (34) becomes

$$\begin{aligned} \mathbb{E}[\mathbf{r}_{\underline{u}}\mathbf{r}_{\underline{u}}^T] &= (1 - r_1) \mathbf{I}_{\kappa} - (1 + r_2) \Lambda_{\kappa} \Lambda_{\kappa}^T \\ &\approx \mathbf{I}_{\kappa} - \Lambda_{\kappa} \Lambda_{\kappa}^T. \end{aligned} \quad (36)$$

By defining e_u as the error generated by DNN optimization $\|\mathbf{r}_{\underline{u}}\|_2^2$, we have

$$\begin{aligned} e_u &:= \|\mathbf{r}_{\underline{u}}\|_2^2 \iff \\ \text{Tr}(\mathbb{E}[\mathbf{r}_{\underline{u}}\mathbf{r}_{\underline{u}}^T]) &\approx \kappa - \text{Tr}(\Lambda_{\kappa} \Lambda_{\kappa}^T). \end{aligned} \quad (37)$$

Based on (4), the symmetric form of $\Sigma_{\underline{u}}^{-1/2} \Sigma_{\underline{u}\underline{y}} \Sigma_{\underline{y}}^{-1/2}$ is derived, that is

$$\begin{aligned} \text{Tr}(\Sigma_{\underline{u}}^{-1/2} \Sigma_{\underline{u}\underline{y}} \Sigma_{\underline{y}}^{-1/2} (\Sigma_{\underline{u}}^{-1/2} \Sigma_{\underline{u}\underline{y}} \Sigma_{\underline{y}}^{-1/2})^T) \\ = \text{Tr}(\Gamma_{\kappa} \Lambda_{\kappa} \Delta_{\kappa}^T \Delta_{\kappa} \Lambda_{\kappa}^T \Gamma_{\kappa}^T) \\ = \text{Tr}(\Gamma_{\kappa} \Lambda_{\kappa} \Lambda_{\kappa}^T \Gamma_{\kappa}^T) \end{aligned}$$

$$= \text{Tr}(\Lambda_{\kappa} \Lambda_{\kappa}^T) \quad (38)$$

that $\text{Tr}(\Gamma_{\kappa} \Lambda_{\kappa} \Lambda_{\kappa}^T \Gamma_{\kappa}^T) = \text{Tr}((\Gamma_{\kappa} \Lambda_{\kappa})^T \Gamma_{\kappa} \Lambda_{\kappa})$ is given by the property of the trace operator. Then, objective function $\mathcal{L}_{\text{corr}}$ in (28) is replaced with

$$\mathcal{L}_{\text{corr}} = \kappa - \text{Tr}(\Lambda_{\kappa} \Lambda_{\kappa}^T). \quad (39)$$

Combining (37) with (39), one has

$$\begin{aligned} \max \rho(\mathbf{u}, \mathbf{y}) &= \max \text{Tr}(\Lambda_{\kappa} \Lambda_{\kappa}^T) \\ \iff \max \Lambda_{\kappa} \Lambda_{\kappa}^T &= \min \text{Tr}(\mathbb{E}[\mathbf{r}_{\underline{u}}\mathbf{r}_{\underline{u}}^T]) = \min \|\mathbf{r}_{\underline{u}}\|_2^2 \end{aligned} \quad (40)$$

and

$$\max \Lambda_{\kappa} \Lambda_{\kappa}^T = \max \sum_{j=1}^{\kappa} \sigma_j^2. \quad (41)$$

Similarly, we can deduce the relationship between $\|\mathbf{r}_{\underline{y}}\|_2^2$ and $\rho(\mathbf{u}, \mathbf{y})$ as

$$\begin{aligned} \max \rho(\mathbf{u}, \mathbf{y}) &= \max \text{Tr}(\Lambda_{\kappa}^T \Lambda_{\kappa}) \\ \iff \max \Lambda_{\kappa}^T \Lambda_{\kappa} &= \min \text{Tr}(\mathbb{E}[\mathbf{r}_{\underline{y}}\mathbf{r}_{\underline{y}}^T]) = \min \|\mathbf{r}_{\underline{y}}\|_2^2. \end{aligned} \quad (42)$$

Hence, optimization $\|\mathbf{r}_{\underline{u}}\|_2^2 + \|\mathbf{r}_{\underline{y}}\|_2^2$ is functional equivalent to maximization $\rho(\mathbf{u}, \mathbf{y})$, this theorem is proved. ■

Remark 4: It is a fact that the optimal operators of the proposed SaECCA method are obtained by minimizing the errors e_u and e_y . The reformulated objective function has the benefits that: 1) it is more practical and robust compared to directly solving $\mathcal{L}_{\text{corr}}$; 2) it facilitates more useful information to be extracted by the DNNs as opposed to relying on correlation alone; 3) it provides a bridge between DNN-aided CCA and nonlinear least squares techniques.

C. SaECCA-Based FD Designs

Fig. 2 sketches the overview flowchart of designed SaECCA algorithm, involving three steps: 1) data collection; 2) offline modeling; and 3) online monitoring. At first, the fault-free data $\mathbf{Z}$ is collected to be used for SaECCA training. With the aid of SKConv, network operators $\mathcal{H}(\Theta_{\mathcal{H}}^*) : U \rightarrow \underline{U}^*$ and $\mathcal{G}(\Theta_{\mathcal{G}}^*) : Y \rightarrow \underline{Y}^*$ are obtained in the offline modeling stage. Based on the optimal parameters $(\Theta_{\mathcal{H}}^*, \Theta_{\mathcal{G}}^*)$ and mapping matrices $(\mathbf{J}_{\mathcal{H}}, \mathbf{L}_{\mathcal{G}})$, the residual vectors in (29) and (30) can be formulated as follows:

$$\mathbf{r}_u = \mathbf{J}_{\mathcal{H}}^T \mathcal{H}(\mathbf{u}, \Theta_{\mathcal{H}}^*) - \Lambda_{\kappa} \mathbf{L}_{\mathcal{G}}^T \mathcal{G}(\mathbf{y}, \Theta_{\mathcal{G}}^*) \quad (43)$$

$$\mathbf{r}_y = \mathbf{L}_{\mathcal{G}}^T \mathcal{G}(\mathbf{y}, \Theta_{\mathcal{G}}^*) - \Lambda_{\kappa} \mathbf{J}_{\mathcal{H}}^T \mathcal{H}(\mathbf{u}, \Theta_{\mathcal{H}}^*). \quad (44)$$

Then, T^2 statistics are constructed for FD

$$T_{r_u}^2 = \mathbf{r}_u^T \Sigma_{r_u}^{-1} \mathbf{r}_u \quad (45)$$

$$T_{r_y}^2 = \mathbf{r}_y^T \Sigma_{r_y}^{-1} \mathbf{r}_y \quad (46)$$

where $\Sigma_{r_u} = \mathbf{I}_{\kappa} - \Lambda_{\kappa} \Lambda_{\kappa}^T$ and $\Sigma_{r_y} = \mathbf{I}_{\kappa} - \Lambda_{\kappa}^T \Lambda_{\kappa}$ denote the covariance of residual vectors $\mathbf{r}_u$ and $\mathbf{r}_y$, respectively.

It should be noted that the feature extraction process of SKConv exhibits inherent nonlinearity, such that its mapping results do not hold Gaussian distribution [17]. To solve this

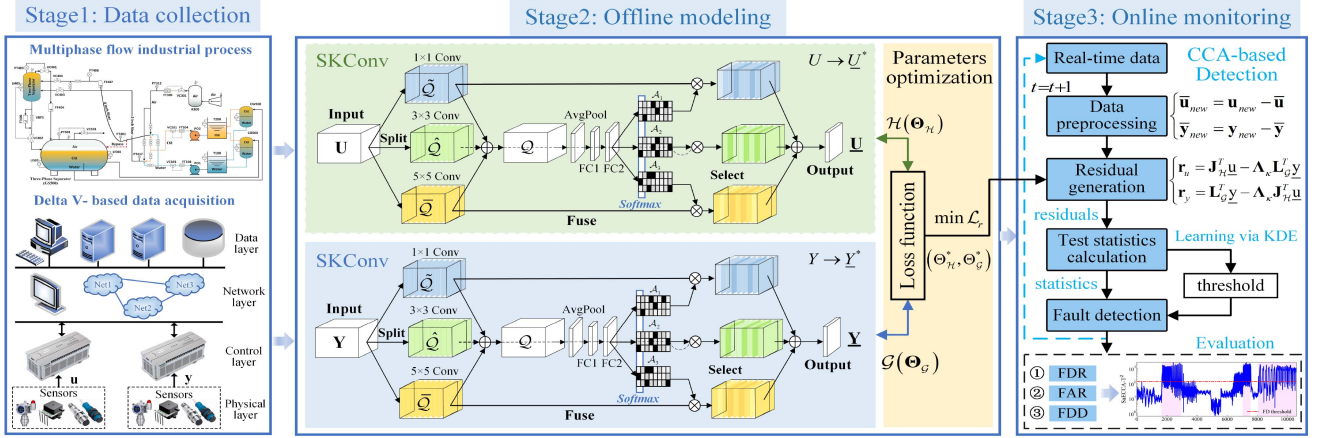


Fig. 2. Overview of the FD procedures for the proposed SaECCA, illustrated by a MPF industrial process.

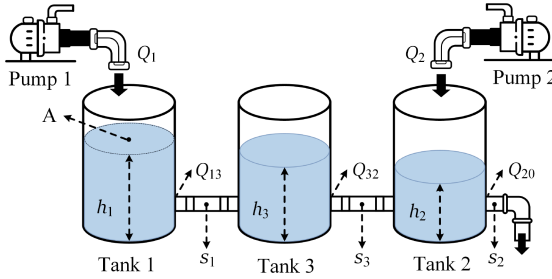


Fig. 3. Schematic diagram of the TTS simulation process.

issue, Kernel density estimation (KDE) [4] is used to estimate the threshold J_{th} of T^2 statistics. For statistics with unknown probability density function $f(x)$, its estimated density at x can be given by

$$\hat{f}(x) = \frac{1}{N_s d} \sum_{i=1}^{N_s} \phi\left(\frac{x - x_i}{d}\right) \quad (47)$$

where x_i is the i th statistic for $i \in \{1, 2, \dots, N_s\}$, and d is the kernel bandwidth, whose selection can be found in literature [32]. Here, the Gaussian kernel function is used

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x^2\right). \quad (48)$$

Under the significant level α , the threshold J_{th} of the T^2 statistics can be calculated:

$$P(T^2 < J_{th}) = \int_{-\infty}^{J_{th}} \hat{f}(T^2) dT^2 = \alpha. \quad (49)$$

Based on (49), the corresponding thresholds J_{th1} and J_{th2} for the statistics $T_{r_u}^2$ and $T_{r_y}^2$ can be estimated. Then, the following FD decision is formulated:

$$H_0 : T_{r_u}^2 \leq J_{th1} \text{ and } T_{r_y}^2 \leq J_{th2} \implies \text{fault-free} \quad (50)$$

$$H_1 : T_{r_u}^2 > J_{th1} \text{ or } T_{r_y}^2 > J_{th2} \implies \text{faulty}. \quad (51)$$

In the online monitoring stage, new samples are acquired in real time, denoted as $\mathbf{u}_{new}$ and $\mathbf{y}_{new}$. After being centralized, the SKR of $\mathbf{u}$ and $\mathbf{y}$ are obtained from the network operators $\mathcal{H}(\Theta_{\mathcal{H}}^*) : \mathbf{U} \rightarrow \mathbf{U}^*$ and $\mathcal{G}(\Theta_{\mathcal{G}}^*) : \mathbf{Y} \rightarrow \mathbf{Y}^*$. Then, the evaluation statistics T^2 are calculated according to (45) and (46). The null hypothesis H_0 corresponds to the system being in the fault-free state, whereas the alternative hypothesis H_1 denotes the system fault, triggering an alarm for the operators.

IV. PARAMETER CONVERGENCE AND DETECTION PERFORMANCE ANALYSIS OF SAECCA

A. Parameter Convergence of SaECCA

Defining $\Theta := (\Theta_{\mathcal{H}}, \Theta_{\mathcal{G}})$, the update rule for the proposed SaECCA parameters can be expressed as

$$\Theta^{k+1} \leftarrow \Theta^k - \eta \nabla \mathcal{L}_r(\Theta^k) \quad (52)$$

where η is the learning rate and k is the current epoch. Further, Theorem 2 is introduced to illustrate that the loss function $\mathcal{L}_r(\Theta)$ can converge to one particular optimal solution.

Theorem 2: For a given (or tolerable) estimation error ϵ , one particular optimal solution $\mathbf{U}^*$ and $\mathbf{Y}^*$ can be obtained after a finite number K of iterations, and $\Theta^K \rightarrow \Theta^*$ with

$$\|\mathcal{L}_r(\Theta^K) - \mathcal{L}_r(\Theta^*)\| \leq \epsilon. \quad (53)$$

Proof: It is clear that operations, such as convolution and soft attention, are Lipschitz continuous. Then, $\mathbf{r}_u$ and $\mathbf{r}_y$ are Lipschitz continuous and bounded with respect to Θ by the chain rule. As such, for $\forall \Theta_1, \Theta_2 \in \Theta$, satisfies

$$\|\nabla \mathcal{L}_r(\Theta_1) - \nabla \mathcal{L}_r(\Theta_2)\| \leq \beta \|\Theta_1 - \Theta_2\| \quad (54)$$

where β is Lipschitz constant for $\nabla \mathcal{L}_r$.

For the loss function $\mathcal{L}_r(\Theta)$, its second-order Taylor expansion about the parameter Θ_2 is given by

$$\begin{aligned} \mathcal{L}_r(\Theta_1) &= \mathcal{L}_r(\Theta_2) + \nabla \mathcal{L}_r(\Theta_2)^T \Delta \Theta \\ &\quad + \frac{1}{2} \mathbf{H}(\Theta_2) (\Delta \Theta)^2 + O(\|\Delta \Theta\|^2) \end{aligned} \quad (55)$$

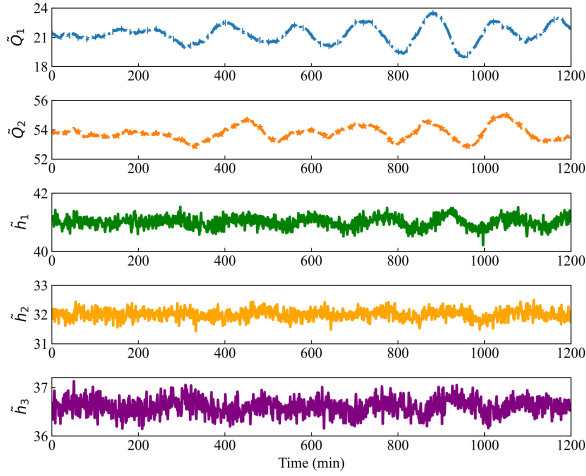


Fig. 4. Process and output variables measured by TTS simulation.

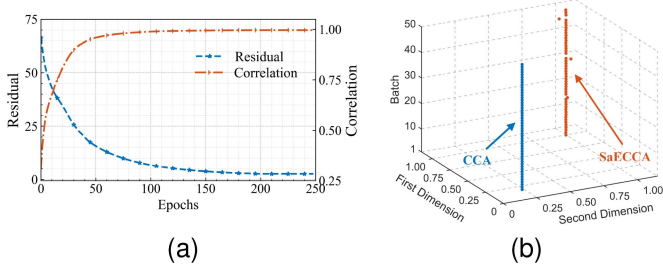


Fig. 5. Training results for TTS simulation. (a) Learning curve in each epoch. (b) Eigenvectors obtained by linear CCA and the SaECCA.

where $\mathbf{H}$ is Hessian matrix and $\Delta\Theta = \Theta_1 - \Theta_2$. Let $\Theta_1 = \Theta^{k+1}$ and $\Theta_2 = \Theta^k$, it yields

$$\mathcal{L}_r(\Theta^{k+1}) \leq \mathcal{L}_r(\Theta^k) + \nabla \mathcal{L}_r(\Theta^k)^T \Delta\Theta + \frac{\beta}{2} \|\Delta\Theta\|_2^2. \quad (56)$$

Then, by integrating the gradient descent update $\Theta^{k+1} = \Theta^k - \eta \nabla \mathcal{L}_r(\Theta^k)$ into (56), we have

$$\mathcal{L}_r(\Theta^{k+1}) \leq \mathcal{L}_r(\Theta^k) - \eta \left(1 - \frac{\beta}{2}\eta\right) \|\nabla \mathcal{L}_r(\Theta^k)\|_2^2. \quad (57)$$

When η is sufficiently small such that $\eta \leq 1/\beta$, it can be derived that

$$\mathcal{L}_r(\Theta^{k+1}) \leq \mathcal{L}_r(\Theta^k) - \frac{\eta}{2} \|\nabla \mathcal{L}_r(\Theta^k)\|_2^2. \quad (58)$$

Using the gradient descent update and (58), the following inequality is given in [33], one has

$$\mathcal{L}_r(\Theta^{k+1}) - \mathcal{L}_r(\Theta^*) \leq \frac{1}{2\eta} D_k \quad (59)$$

where $D_k = \|\Theta^k - \Theta^*\|_2^2 - \|\Theta^{k+1} - \Theta^*\|_2^2$. Then, by repeated iteration and summation up to K , (59) becomes

$$\sum_{k=1}^K \mathcal{L}_r(\Theta^k) - K\mathcal{L}_r(\Theta^*) \leq \frac{1}{2\eta} \sum_{k=1}^K D_k. \quad (60)$$

Note $\sum_{k=1}^K D_k = \|\Theta^1 - \Theta^*\|_2^2 - \|\Theta^K - \Theta^*\|_2^2$. With a simple manipulation, the inequality in (60) becomes

$$\mathcal{L}_r(\Theta^K) - \mathcal{L}_r(\Theta^*) \leq \frac{1}{2\eta K} \|\Theta^1 - \Theta^*\|_2^2. \quad (61)$$

From (61), it can be concluded that there exists a small boundary ϵ such that (53) holds after a finite number K of iterations. Then, this theorem is proven. ■

Remark 5: It is worth noting that: 1) the proposed SaECCA converges with a rate $O(1/K)$ after K iterations; and 2) the optimal solution of SaECCA is unique by transforming the correlation into Euclidean distance.

B. Detection Performance for Incipient Faults

Fault detectability represents the ability to identify the presence of a fault [28]. Combining (10) and (11), the residuals signals with fault vectors $\Xi_u f_u$ or $\Xi_y f_y$ are:

$$\mathbf{r} = \mathbf{r}_n + \mathbf{r}_f. \quad (62)$$

In general, $\mathbf{r}_f$ is uncorrelated with both $\mathbf{u}$ and $\mathbf{y}$. Then, the statistics T^2 under a fault condition can be rewritten as

$$\begin{aligned} \mathbb{E}[T_f^2] &= \mathbb{E}[\mathbf{r}^T \Sigma_r^{-1} \mathbf{r}] \\ &= \mathbb{E}[(\mathbf{r}_n + \mathbf{r}_f)^T \Sigma_r^{-1} (\mathbf{r}_n + \mathbf{r}_f)] \\ &= \mathbb{E}[T^2] + \mathbb{E}[\mathbf{r}_f^T (\mathbf{I}_\kappa - \Lambda_\kappa \Lambda_\kappa^T)^{-1} \mathbf{r}_f] \\ &= T^2 + \mathbf{r}_f^T \text{diag}\left(\frac{1}{1 - \sigma_1^2}, \dots, \frac{1}{1 - \sigma_\kappa^2}\right) \mathbf{r}_f. \end{aligned} \quad (63)$$

As proven in Theorem 1, the optimal network operators $\mathcal{H}(\Theta_{\mathcal{H}}^*) : U \rightarrow U^*$ and $\mathcal{G}(\Theta_{\mathcal{G}}^*) : Y \rightarrow Y^*$ maximize eigenvalues from σ_1 to σ_κ by optimizing objective function $\mathcal{L}_r$ in (32). It means that $\text{diag}(1/(1 - \sigma_1^2), \dots, 1/(1 - \sigma_\kappa^2))$ will be maximized, that is, the fault term in $\mathbb{E}[T_f^2]$ reaches its maximum. Thus, the proposed method can amplify the impact of incipient faults on the statistics T^2 to the maximum extent and achieve optimal FD performance.

V. CASE STUDIES

A. Case Study I: TTS Simulation

1) *Dataset Description:* The TTS introduced by Ding in [2] is widely served as a benchmark process for FD due to its typical chemical industry characteristics. As illustrated in Fig. 3, the TTS benchmark process comprises interconnected tanks, pipelines, and pumps. Based on Torricelli's law and the parameters given in [2], the dynamic equations of TTS with fault signals are formulated as follows:

$$\begin{aligned} A\dot{h}_1 &= Q_1 + f_1 + f_2 - Q_{13} - f_6 \check{Q}_{10} + \xi_1 \\ A\dot{h}_2 &= Q_2 + f_1 + f_3 + Q_{32} - f_6(f_8 + 1)\check{Q}_{20} + \xi_2 \\ A\dot{h}_3 &= Q_{13} - Q_{32} - f_6 \check{Q}_{30} + \xi_3 \\ Q_{13} &= a_1 s_{13}(f_7 + 1)\text{sign}(h_1 - h_3)\sqrt{2g|h_1 - h_3|} \\ Q_{32} &= a_3 s_{23}(f_9 + 1)\text{sign}(h_3 - h_2)\sqrt{2g|h_3 - h_2|} \\ \check{Q}_{i0} &= a_i s_0 \sqrt{2gh_i}, \quad i = 1, 2, 3 \end{aligned} \quad (64)$$

where Q_1 and Q_2 are incoming mass flow (cm^3/s); Q_{ij} is the mass flow from tank i to tank j ; $\check{Q}_1, \check{Q}_2, \check{h}_1, \check{h}_2$, and $\check{h}_3$ are the variables measured by sensors; ξ_1, ξ_2 , and ξ_3 refer to process noises. The remaining parameters can be found in [2].

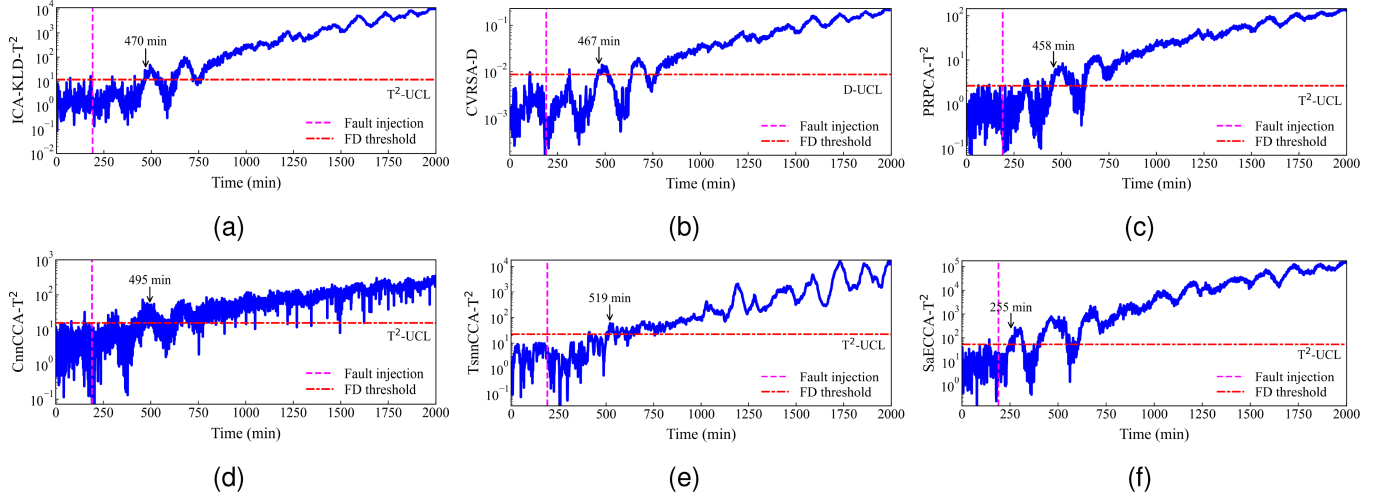


Fig. 6. FD results for fault F7 on TTS simulation. (a) ICA-KLD. (b) CVRSA. (c) PRPCA. (d) CnnCCA. (e) TsnmCCA. (f) SaECCA.

TABLE I
FAULT CASES IN THE TTS SIMULATION

Cases	Fault description	Location
F1	$f_1(t) = 6 \sin\left(\frac{t-200}{6\pi}\right)$	Q_{11}, Q_{22}
F2	$f_2(t) = 0.02(t - 200)$	Q_{11}
F3	$f_3(t) = \frac{t-200}{200} + 5\left(1 - \left[\frac{t}{48} - \frac{5}{4}\left[\frac{t}{60}\right]\right]\right)$	Q_{22}
F4	$f_4(t) = 0.005(t - 200)$	$\tilde{h}_2$
F5	$f_5(t) = 0.008(t - 200)$	$\tilde{h}_3$
F6	$f_6(t) = 0.00005(t - 200)$	$\tilde{Q}_{10}$
F7	$f_7(t) = -0.0002(t - 200)$	Q_{13}
F8	$f_8(t) = -0.0003(t - 200)$	$\tilde{Q}_{20}$
F9	$f_9(t) = -0.0005(t - 200)$	Q_{32}

As indicated in [27], a fault-free dataset with 16 008 samples and nine faulty datasets with 2001 samples are collected with a sampling interval of 1 min. Fig. 4 illustrates the process and output variable measurements in the fault-free dataset, while Table I provides a detailed description of the corresponding incipient fault cases.

2) Parameter Configuration: To verify the superiority of the proposed SaECCA method for incipient FD purpose, five comparison methods, including: ICA-KLD [22], CVRSA [23], PRPCA [24], CnnCCA [20], and TsnmCCA [21], are implemented. The parameters and network structures of the comparison methods are given as follows.

- 1) ICA-KLD: The number of independent components is set to 4 and the basis function is chosen as a Gaussian kernel function.
- 2) CVRSA: The past and future time lags are 5, and the canonical variable dimension is chosen as 4.
- 3) PRPCA: The number of principal components is set to 4, and p_b and p_e are 2 and 2.5, respectively.
- 4) CnnCCA: 1D-Conv (64 cells) $\Rightarrow$ 1D-Conv (128 cells) $\Rightarrow$ Flatten $\Rightarrow$ Dense (32 cells) $\Rightarrow$ Dense (3 cells).

- 5) TsnmCCA: 1) spatial flow: 1D-Conv (64 cells) $\Rightarrow$ 1D-Conv (64 cells) $\Rightarrow$ Pool $\Rightarrow$ Dense (3 cells); 2) temporal flow: Long Short-Term Memory (LSTM, 64 units) $\Rightarrow$ LSTM (64 cells) $\Rightarrow$ Pool $\Rightarrow$ Dense (3 cells).

To determine the FD threshold, the confidence level of $\alpha=0.99$ is used, which indicates that the false alarm rate (FAR) of the statistics is stable around 1%. In addition, by computing the autocorrelation function of all measurements [34], the optimal window width $w=12$.

3) Experimental Results: The loss function of the proposed SaECCA method and the correlation of the nonlinear mapping in the training process are depicted in Fig. 5(a). It is observed that the mean-squared errors of the residual generator decrease gradually with the number of epochs. After around 250 epochs, $\mathcal{L}_r$ reaches convergence, and the correlation between $(\underline{u}, \underline{y})$ becomes the maximum, close to 1. The proofs of Theorems 1 and 2 are completed.

In fact, Fig. 5(a) reveals that the loss function $\mathcal{L}_r$ is closely related to the correlation embedded within the inputs and outputs. Moreover, the correlation can also be measured using the eigenvalues obtained from SVD, as illustrated in Fig. 5(b). It can be observed that the eigenvalues obtained by the SaECCA (orange points) are larger than that obtained by the linear CCA (blue points). This indicates that the SKRs mapped by network operators $\mathcal{H}(\Theta_{\mathcal{H}}^*) : U \rightarrow \underline{U}^*$ and $\mathcal{G}(\Theta_{\mathcal{G}}^*) : Y \rightarrow \underline{Y}^*$ enhance the data correlation. Combined with (63), the proposed SaECCA method can amplify the effect of incipient faults to the greatest extent, thereby enhancing its effectiveness in detecting such faults.

The detection results of six methods for fault F7 on TTS simulation are shown in Fig. 6. Among them, the FD thresholds for each method is determined by the KDE, and fault detection delay (FDD) denotes the number of samples required for the test statistics to exceed the threshold five consecutive times after the fault occurs. It is easy observed that the proposed method can separate dominant variations caused by incipient faults from the normal states. As well, the SaECCA exhibits the lowest

TABLE II
PERFORMANCE COMPARISON FOR THE TTS FAULTS

Faults	ICA-KLD		CVRSA		PRPCA		CnnCCA		TsnmCCA		SaECCA	
	FDR	FAR	FDR	FAR	FDR	FAR	FDR	FAR	FDR	FAR	FDR	FAR
F1	96.83	0.50	95.83	0.52	95.94	0.50	97.72	1.02	94.21	1.02	98.78	0.50
F2	93.83	1.54	93.72	1.52	90.83	1.00	95.94	0.51	94.21	0	97.49	1.04
F3	97.66	0.50	98.39	0.51	98.50	1.00	98.55	0.50	98.94	1.04	98.99	0.50
F4	95.44	0.50	96.83	0	95.39	1.00	87.77	0.50	95.44	0	96.44	0
F5	92.11	0	93.17	0	96.27	0	95.44	1.50	98.11	0	96.16	0
F6	89.44	0.50	88.28	0.55	86.61	0.50	90.88	0	93.77	1.04	95.88	0
F7	78.61	1.01	75.11	1.04	83.45	1.00	81.39	1.51	80.28	0	89.94	0.50
F8	96.94	0.50	97.33	0	97.94	0	96.11	0.50	97.10	0	98.22	0
F9	87.66	0	86.94	1.05	85.95	0.50	88.88	1.04	89.66	1.00	87.38	0

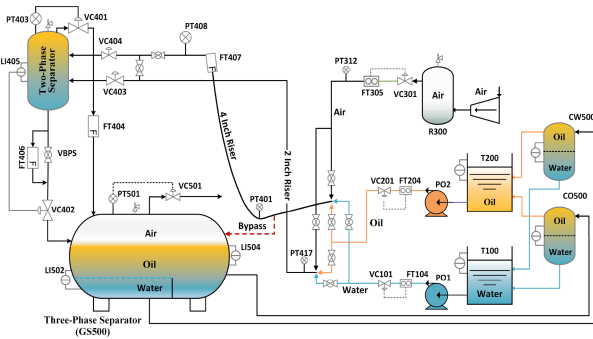


Fig. 7. Schematic diagram of the MPF industrial process.

TABLE III
FAULT CASES IN MPF INDUSTRIAL PROCESS

Cases	Fault description	Operating states	Type
F1	Air line blockage	Steady-state	Incipient
F2	Water line blockage	Changing	Incipient
F3	Top separator input blockage	Changing	Incipient
F4	Open direct bypass	Steady-state	Incipient
F5	Slugging conditions	Changing	Intermittent
F6	Pressurization of the 2" line	Changing	Abrupt

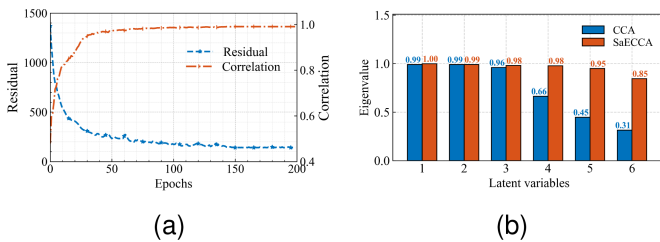


Fig. 8. Training results for MPF industrial process. (a) Learning curve in each epoch. (b) Latent variable eigenvalues in CCA and SaECCA.

FDD, which detect the incipient faults earlier compared to other methods. This indicates that the SaECCA can effectively amplify the incipient fault information and achieve satisfactory FD performance by using the weighting strategy with the self-attention sliding window.

To demonstrate the superiority of the proposed SaECCA method for FD purpose, two key evaluation indicators, such as FAR, and fault detection rate (FDR), are summarized in Table II. As seen from Table II, SaECCA exhibits the highest FDR and the lowest FAR for most of the fault scenarios compared to the other five methods. This result indicates that SaECCA can accurately identify these faults, which verified the effectiveness of the DNN-aided CCA methods for FD task. It can also be noted that TsnmCCA achieves the best FD performance for some fault scenarios, such as fault F5 and F9. This is because TsnmCCA

extracts spatio-temporal features from the data through a spatial stream (CNN) and a sequential stream (LSTM), which makes it to more effectively capture long-term temporal dependencies. However, compared to TsnmCCA, SaECCA employs a residual-based optimization strategy and exhibits superior false alarm suppression, yielding the lowest average FAR. Hence, the proposed method demonstrates more robustness and can provide a more reliable and accurate fault warning result for operators.

B. Case Study II: MPF Industrial Benchmark

1) Dataset Description: The Cranfield University has designed a MPF facility for simulating real pressure systems under different operating conditions. Fig. 7 presents the sketch of the MPF industrial process. The entire facility primarily includes: a gas and liquid two-phase separator, a three-phase separator (GS500), coalescers (CW500 and CO500), storage tanks (T100 and T200), air compressors, multistage pumps (PO1 and PO2), and pipelines and valves of various sizes and configurations. The facility controls the pressure of the pressurized system by adjusting the setpoints of the air and water flowrates with typical dynamic characteristics. More details can be found in [34].

The system has collected 24 measured variables at a sampling rate of 1 Hz. To better reflect the complexity of dynamic industrial process, three normal datasets with 20 distinct combinations of air and liquid flowrates are obtained from the SCADA system. There are six fault conditions, including blockage, slugging, and others, described in Table III. It is worth noting that variable

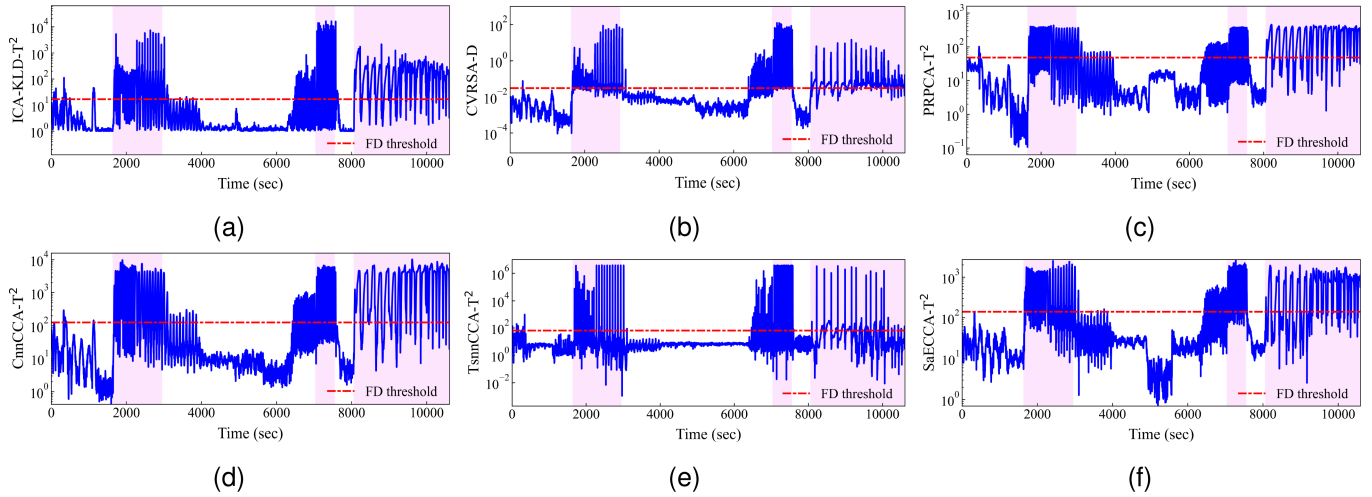


Fig. 9. FD results for fault F5 on MPF industrial process. (a) ICA-KLD. (b) CVRSA. (c) PRPCA. (d) CnnCCA. (e) TsmnCCA. (f) SaECCA.

24 was only used for fault F6, while the remaining fault cases utilized the first 23 variables for algorithm training and analysis.

2) Parameter Configuration: The optimal window width w is determined to be 15, and parameter configuration for the comparison methods are provided as follows.

- 1) ICA-KLD: The number of independent components is set to 6 and the basis function is chosen as a Gaussian kernel function.
- 2) CVRSA: The past and future time lags are 3, and the canonical variable dimension is chosen as 10.
- 3) PRPCA: The number of principal components is set to 6, and p_b and p_e are 0.5 and 1, respectively.
- 4) CnnCCA: 1D-Conv (32 cells) $\Rightarrow$ 1D-Conv (32 cells) $\Rightarrow$ Flatten $\Rightarrow$ Dense (10 cells).
- 5) TsmnCCA: 1) spatial flow: 1D-Conv (32 cells) $\Rightarrow$ 1D-Conv (32 cells) $\Rightarrow$ Pool $\Rightarrow$ Dense (8 cells); 2) temporal flow: LSTM (32 cells) $\Rightarrow$ LSTM (32 cells) $\Rightarrow$ Pool $\Rightarrow$ Dense (8 cells).

3) Experimental Results: Base on the MPF data, the residual variations and correlations of the proposed SaECCA method during the training are shown in Fig. 8(a). In this experiment, the SaECCA is more difficult to train compared with the TTS simulation. This is due to the fact that dataset from MPF industrial process is subject to frequent operating condition variations, and contain more measured variables and random noise. It can lead to a larger estimation error $e_n = \mathbb{E}[\mathcal{L}_r(\Theta^*)] - \inf \mathcal{L}_r$, and prevent the correlation between the $(\mathbf{u}, \mathbf{y})$ to stably converge to 1. Note that $\inf \mathcal{L}_r$ is the boundary of DNNs learning ability and e_n is the bias-variance error. As can be seen from Fig. 8(b), the eigenvalues for the first three dimensions of both methods achieve the maximum correlation, close to 1. However, as the latent variable dimensions increase, the eigenvalues obtained by linear CCA become progressively smaller and fail to maintain the maximum correlation, while the eigenvalues obtained by SaECCA remain essentially unchanged.

The FD results for fault F5 in the MPF industrial process are given in Fig. 9, in which the pink background denotes the fault area. It can be seen that the six methods can successfully detect

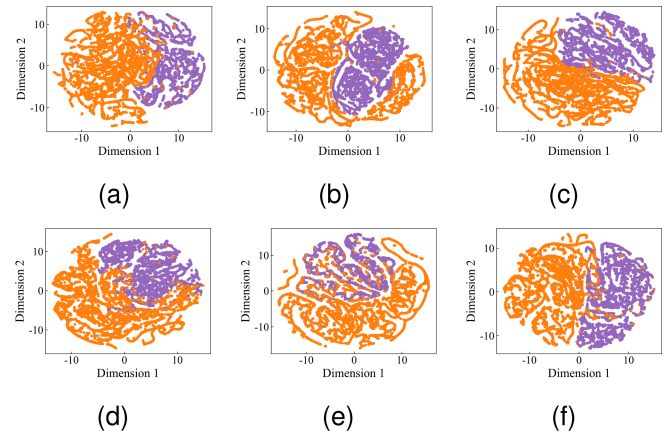


Fig. 10. t-SNE visualizations for fault F5. (a) ICA-KLD. (b) CVRSA. (c) PRPCA. (d) CnnCCA. (e) TsmnCCA. (f) SaECCA.

the fault F5, but the proposed method extracts more distinct fault features with better false alarm suppression. To further validate the superiority of SaECCA, Fig. 10 visualizes the statistical vectors using t-distributed stochastic neighbor embedding (t-SNE), purple and orange points are normal and fault data, respectively. As can be seen in Fig. 10, the SaECCA obtains a clearer classification boundary and can more effectively distinguish between system states compared to other FD methods. The evaluation results of various methods are presented in Table IV.

From the evaluation results, it can be concluded that the SaECCA obtains the highest FDR and the lowest FAR compared to DNN-aided CCA methods. This verifies the effectiveness of the self-attention sliding window strategy, and the robustness of the minimization residual-based optimization. From Table IV, it can be observed that the FAR for all methods deviates from the expected level in faults F1, F4, and F5. It is because faults F1 and F4 are progressive faults, where the statistics do not drop rapidly but instead gradually decrease below the control limit once the

TABLE IV
PERFORMANCE COMPARISON FOR THE MPF FAULTS

Faults	ICA-KLD		CVRSA		PRPCA		CnnCCA		TsnmCCA		SaECCA	
	FDR	FAR	FDR	FAR	FDR	FAR	FDR	FAR	FDR	FAR	FDR	FAR
F1	79.96	4.23	77.29	6.41	71.15	5.15	77.37	6.47	80.41	3.69	80.11	3.13
F2	82.33	5.67	76.28	5.65	82.89	3.29	80.11	2.96	80.83	1.89	83.38	1.72
F3	98.68	1.71	94.43	0.86	97.27	1.22	99.14	1.23	98.65	2.88	99.20	1.12
F4	80.52	5.42	81.79	5.49	83.46	5.61	80.93	6.97	81.50	5.66	88.05	5.22
F5	77.45	6.52	79.40	5.20	71.27	7.51	76.17	7.53	78.36	6.15	82.88	5.89
F6	99.63	0	99.81	0	99.81	0	99.91	0.17	99.81	0	99.91	0

fault stops being injected. Similarly, the same phenomenon can be observed in the intermittent fault F5.

VI. CONCLUSION

In this article, an SaECCA method with the aid of DNNs, which can effectively parameterize the SKR of the residual generators, has been proposed for the incipient FD within dynamic industrial processes. By dynamically learning temporal correlations hidden in nonlinear systems using the self-attention sliding window, the method overcomes the limitations of static weighting strategies so that the influence of incipient faults is maximized to the best extent. As shown by the theoretical proofs and mathematical analysis, the proposed SaECCA method delivers the optimal FD performance. Experimental results illustrate that this scheme can effectively distinguish the system states, and improve the detection ability for incipient faults. The considerable contribution of this work lies in providing a novel and robust optimization scheme for DNN-aided CCA methods. Besides, to achieve optimal FD performance, the conventional CCA method requires that the two groups of variables follow a joint Gaussian distribution. However, existing DNN-CCA methods are not subject to such constraints, so the distribution of the mapping vectors can be further adjusted using generative DNN models.

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